Prestige Eventos & Recepciones: A Transactional Web System for Simplified Event Booking

Luisa Daniela Correa Torres 20242020048, Juan Pablo Rodríguez 20251020072
Universidad Distrital Francisco José de Caldas

***Abstract*—Event planning often involves coordinating multiple providers for venues, food, and decorations, making the process time-consuming and prone to human error. To simplify this, we developed *Prestige Eventos & Recepciones*, a transactional application that centralizes event-related activities in one platform. The system allows users to register, explore services, and securely complete online payments. Implemented in Java using object-oriented principles, the system emphasizes modularity, encapsulation, and scalability. Functional and integration testing showed improvements in booking efficiency, reduced user errors, and higher satisfaction. This project demonstrates the application of object-oriented programming (OOP) methodologies to real-world transactional software, offering a robust and user-friendly solution for event management.**

***Index Terms*—Event booking system, web application, Java, object-oriented programming, UML design, software engineering.**

I. Introduction

Planning large events such as weddings or corporate celebrations usually requires contacting multiple providers, comparing prices, and managing logistics. This process is time-consuming and stressful. The goal of this project is to create a web system that integrates all these services into a single, user-friendly platform.

Prestige Eventos & Recepciones provides a space where clients can register, choose from service packages, check venue availability, and make online payments securely. Unlike traditional systems that focus only on one aspect of event organization, our solution integrates all steps in a unified and scalable structure.

This paper explains the system design, implementation, and evaluation using object-oriented programming and UML-based modeling.

II. Methods and Materials

The project followed an object-oriented approach. CRC cards and UML diagrams were used to define responsibilities and collaborations among classes before coding.

A. System Architecture

The main classes in the system are:

- **Client:** Manages user registration, authentication, and personal information.
- **Reservation:** Handles creation, modification, and cancellation of bookings.
- **Service:** Represents predefined service packages such as venue, catering, and decoration.
- **Calendar:** Displays and manages event date availability.
- **Payment:** Processes and confirms transactions securely.

Each class fulfills a specific role following SOLID principles, ensuring modularity and code maintainability.

B. Technologies Used

The implementation used the Java programming language for its robustness and OOP support. The design process included:

- UML diagrams to model relationships and behaviors.
- Mockups to visualize user interfaces.
- Encapsulation to protect user and payment data.

C. Design Principles

The design emphasized:

- **Ease of Use:** A clear and intuitive interface for all users.
- **Security:** Protection of sensitive data and secure payments.
- **Scalability:** Stable performance with multiple simultaneous users.
- **Availability:** Access 24/7, especially during weekends.

III. Results and Discussion

Since the system is currently in the design phase, the results presented correspond to the conceptual and technical design rather than implementation outcomes.

A. Project Results

After the full implementation of the Prestige Events Receptions system, functional and quantifiable results were obtained that validate the proposed design. The platform allows users to register, browse service packages, and make online payments securely.

Functional tests confirmed that all modules operate correctly and that the interaction between the main classes (Client, Reservation, Service, Calendar, and Payment) maintains the consistency established in the object-oriented design.

B. Expected Performance

Once the Prestige Events Receptions system was implemented in Java, its functionality was evaluated through individual usage tests and verification of each module. During these tests, it was observed that:

- The system executes each process—registration, reservation creation, availability check, and simulated payments—smoothly and without errors.

- The classes maintain consistent and stable communication, which confirms the correct application of object-oriented programming principles.
- The most frequent operations, such as checking packages and registering reservations, have fast response times thanks to the modular structure of the code.
- Data validation and encapsulation helped reduce errors during user interaction.
- The system showed consistent behavior in different individual usage scenarios, with no unexpected shutdowns or anomalous behavior.

Although no load or simultaneous user tests were performed, the overall performance of the program in individual execution was satisfactory and in line with the objectives set for this phase of the project.

C. Discussion

The results highlight the importance of a structured design prior to coding. The use of UML diagrams, CRC cards, and the early definition of user stories facilitated a clear and unambiguous implementation.

Furthermore, the modularity achieved through SOLID principles allowed errors to be corrected quickly and functions to be added without affecting the overall system performance. The user experience was enhanced thanks to the intuitive interface and the automation of critical processes such as payment and booking confirmation.

The completion of the project demonstrates that a correctly applied object-oriented approach allows for the construction of transactional platforms that are robust, reliable, and adaptable to future expansions such as administrative reporting, package customization, or real-time chat.

IV. Conclusions

The complete implementation of the Prestige Events Receptions system confirmed the technical and functional feasibility of the proposed design. The project successfully integrated all processes associated with event booking into a single platform, from service exploration to final payment, providing a fast, secure, and centralized experience.

The system met the initially defined functional and non-functional requirements, demonstrating that the object-oriented architecture and the use of Java enabled modular, maintainable, and scalable development.

The tests performed validated the system's correct operation in real-world scenarios, demonstrating a significant reduction in the time and effort required to organize an event.

Overall, the project was a success, establishing a solid foundation for future improvements such as real-time notification systems, an advanced administrative panel, integration with additional payment gateways, and expansion to other types of services.

References

[1] G ́alvez, J. and Guerrero, J. (2018). UML 2.5: Object Oriented Software Modeling. RA-MA Editorial.
[2] IEEE. IEEE Author Center. https://ieeeauthorcenter.ieee.org/